World's Longest Bridge Over Water

The bridge spans Jiaozhou Bay, on the sudden coast of Chinas Shandong peninsula in north eastern China. It travels 26.4 miles long. Chinese workers finished this construction in four years, starting at each side and meeting in middle. The structure has 5,200 pillars and cost 2.3 billion dollars.



(i) Advantages of SCC

- No need of mechanical vibrators.
- Faster construction improve working conditions.
- Greater freedom in design.
- Less noise due to absence of mechanical vibrators.
- Improves the quality, durability and reliability of concrete.

2.2.5 Ferrocement

Ferrocement is defined as a thin walled reinforced concrete. It is reinforced with small diameter wire mesh instead of steel rods. Wire mesh is usually made of steel 0.8 to 1.0 mm diameter with 5 to 50 mm spacing. The CM generally used is in ratio of 1:2 or 1:3.



(i) Properties of ferrocement

1. It has high strength per unit mass.
2. It has good capacity to resist shock.
3. Highly versatile form of reinforced concrete.
4. It does not need form work.
5. It is impervious.

(ii) Uses of ferrocement

1. Partition wall.
2. Window frames, shutters, sunshades, etc.
3. Cupboard shelves.
4. Precast roof elements.
5. Domestic water tank.
6. Furniture.
7. Manhole covers.
8. Boats.

2.3 Glass

Glass is a hard, brittle, transparent or translucent material. It is used in numerous applications in our daily lives. Glass is made from natural and abundant raw materials (sand, soda ash and lime stone) that are melted at very high temperature.

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